Henry Phan

North York, ON | henry.phan@torontomu.ca | (437) 928-8314

Summary of Qualifications

- Highly motivated 4th-year Computer Engineering student with strong engineering analysis and report-writing skills developed through several years of innovative academic projects.
- Hands-on experience in assembling, testing, and troubleshooting embedded systems and microcontrollers, with a strong foundation in system-level architecture.
- Working proficiency in test automation and data management utilizing Python, alongside a strong ability to interpret and analyze system-level schematics.

Technical Skills

Languages: C, C++, Java, Python, MATLAB, VHDL, HCS12 Assembly, SQL
Hardware: Altera FPGA, Arduino Uno, HCS12, Breadboards, Oscilloscopes, Digital Multimeters
Software/Tools: Linux, AutoCAD, Power BI, Microsoft Office (Excel, PowerPoint, Word), Quartus II

Education

Toronto Metropolitan University — Toronto, ON
Bachelor of Engineering, Computer Engineering — *2023 – Present*

- Anticipated 2028

Experience

Head Counter — April 2022 – September 2023
Sugarhill Desserts

- Proposed seasonal promotional ideas to the manager to attract customers.
- Debugged POS system errors to restore payment processing and sales reporting functionality.
- Assisted in training new staff members in a fast-paced environment.
- Performed inventory stock procedures to maintain inventory levels at all times.

Projects

CPU Design — November 2024

- Designed and implemented a VHDL-based CPU on an Altera FPGA board to execute arithmetic and logic operations.
- Utilized block diagrams and code to integrate components such as an ALU, D flip-flops, 3-to-8 decoders, encoders, and SSEGs.
- Evaluated system performance through waveform simulations and hardware testing across all implemented operations.

Embedded Navigation Robot — November 2025

- Programmed an HCS12 microcontroller in Assembly with a team to autonomously navigate a maze, detecting roadblocks and intersections.
- Implemented sensor-scanning, motor-control, and navigation logic to process photodetector and bumper inputs for real-time robot control.
- Utilized interrupts, hardware registers, state-machine logic, and subroutines to implement time-critical navigation routines.

Reminder Bot — June 2026

- Created a Python Discord bot using `discord.py` and SQLite, allowing users to create and manage reminders for major upcoming events via chat.
- Designed a reminder system to set up automated countdown alerts and notifications directly to specific user accounts.
- Implemented exception handling and input validation to manage invalid user inputs and prevent application crashes.